Periodic Solutions and Chaos in a Non-Abelian Coil

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Outline

1. The setup
2. Time dependent ansatz
3. Analytic solutions
4. Exploring the phase space beyond the integrable sector
5. The probe scalar field
6. Further remarks

Motivations

- Time-periodic configurations in nonlinear hyperbolic problems are challenging to construct but are of significant physical interest.
- We will study Yang-Mills for the $SU(2)$ gauge group coupled minimally to a real scalar in the adjoint with a Higgs-like self interaction potential.
- The construction of time-dependent configurations offer an opportunity to explore chaotic behavior in Yang-Mills theory, which has been of significant interest in recent years.

The setup

The setup

The theory that we will study is the Georgi-Glashow model: Yang-Mills $SU(2)$ minimally coupled to a real scalar in the adjoint with fixed background spacetime

$$S = \int \mathrm{d}^4x\sqrt{-g}\left[-\frac{1}{4e^2}F^a_{\mu\nu}F^{a\mu\nu} - \frac{1}{2e^2}D_\mu\Phi^a D^\mu\Phi^a - \frac{\lambda}{4}\left(\Phi^a\Phi^a - \nu^2\right)^2\right]$$

The field strength and covariant derivatives are

$$\begin{aligned} F &= \mathrm{d}A + A\wedge A \qquad D_\mu = \nabla_\mu + [A_\mu, \cdot] \\ A &= A^a_\mu dx^\mu t_a \ , \qquad \Phi = \Phi^a t_a \ , \\ t_a &= i\sigma_a \ . \end{aligned}$$

The equations are

$$\begin{aligned} D_\mu F^{\mu\nu} - [\Phi, D^\nu\Phi] &= 0 \ , \\ D_\mu D^\mu\Phi - e^2\lambda\left(\Phi^a\Phi_a - \nu\right)\Phi &= 0 \ . \end{aligned}$$

The setup

With differential forms the action is

$$
\begin{aligned}
S &= \int -\frac{1}{2e^2} F^a \wedge \star F^a - \frac{1}{2e^2} D\Phi^a \wedge \star D\Phi^a - V(\Phi) \star 1 \ , \\
D\Phi &= d\Phi + [A, \Phi] \ .
\end{aligned}
$$

and the equations

$$
\begin{aligned}
-d \star F - A \wedge \star F + \star F \wedge A - [\Phi, \star D\Phi] &= 0 \ , \\
d \star D\Phi + A \wedge \star D\Phi + \star D\Phi \wedge A - \lambda \left(\Phi^a \Phi^a - \nu^2 \right) \star 1 &= 0 \ .
\end{aligned}
$$

The setup

The energy momentum tensor of the configuration

$$\begin{aligned} T_{\mu\nu} &= D_\mu \Phi^a D_\nu \Phi_a - \frac{1}{2} g_{\mu\nu} D_\sigma \Phi^a D_\sigma \Phi_a - g_{\mu\nu} \frac{\lambda e^2}{4} \left(\Phi^a \Phi_a - \nu^2\right)^2 \\ &\quad + F^a{}_{\mu\lambda} F_{a\nu}{}^{\lambda} - \frac{1}{4} g_{\mu\nu} F_{a\rho\sigma} F^{a\rho\sigma} \end{aligned}$$

and the energy density is $\mathcal{E} = T_{00}$.

The presence of a scalar field in the adjoint allows us to construct a gauge invariant quantity

$$F_{\text{eff}} = \mathrm{tr}\left(\Phi F\right)$$

$F \to U^{-1} F U \ , \qquad \Phi \to U^{-1} \Phi U$
$F_{\text{eff}} \to \mathrm{tr}\big(U^{-1} \Phi U U^{-1} F U\big) = \mathrm{tr}\big(U^{-1} \Phi F U\big) = \mathrm{tr}(\Phi F) = F_{\text{eff}}$

Time dependent ansatz

The time-dependent ansatz

Let us consider flat spacetime in cylindrical coordinates

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}z^2 + \mathrm{d}\rho^2 + \rho^2 \mathrm{d}\varphi^2$$

the range of the coordinates are as usual $\rho \in [0, +\infty[$ $t, z \in \mathbb{R}$ and $\varphi \sim \varphi + 2\pi$. In this background we consider the following ansatz

$$\begin{aligned} A &= -\frac{W(t)}{\sqrt{2}} \left(\mathrm{t}_1 \rho \mathrm{d}\varphi - \mathrm{t}_2 \mathrm{d}\rho \right) - \frac{1}{2} \mathrm{t}_3 \mathrm{d}\varphi \ , \\ \Phi &= G(t)\, \mathrm{t}_3 \ . \end{aligned}$$

From now on $\frac{\mathrm{d}X}{\mathrm{d}t} = \dot{X}$.

The time-dependent ansatz

In the ansatz the non-Abelian field strength is

$$F = \frac{\dot{W}}{\sqrt{2}} \left(\mathrm{d}t \wedge \mathrm{d}\rho \mathrm{t}_2 - \rho \mathrm{d}t \wedge \mathrm{d}\varphi \mathrm{t}_1 \right) - W^2 \rho \mathrm{d}\rho \wedge \mathrm{d}\varphi \mathrm{t}_3$$

and the effective abelian field strength is

$$F_{\mathsf{eff}} = \mathrm{tr}\left(F\Phi \right) = \rho G W^2 \mathrm{d}\rho \wedge \mathrm{d}\varphi$$

The time-dependent ansatz

In the ansatz the energy momentum has natural cylindrical symmetry

$$T_{\mu\nu}\mathrm{d}x^\mu \otimes \mathrm{d}x^\nu = \mathcal{E}\mathrm{d}t^2 - p_\perp\left(\mathrm{d}\rho^2 + \rho^2\mathrm{d}\theta^2\right) - p_z\mathrm{d}z^2 \ ,$$

where

$$\begin{aligned}
\mathcal{E} &= \frac{1}{2}\left(\dot{G} + \dot{W}^2\right) + 2W^2G^2 + \frac{1}{2}W^4 + \frac{\lambda}{4}\left(G^2 - \nu^2\right)^2 \\
p_\perp &= -\frac{1}{2}\dot{G} - \frac{1}{2}W^4 + \frac{\lambda}{4}\left(G^2 - \nu^2\right)^2 \\
p_z &= p_\perp + 2W^2G^2 + W^4 - \frac{1}{2}\dot{W}^2
\end{aligned}$$

Easy way to evaluate the equation with the computer

Recall that the action is

$$S = \int -\frac{1}{2e^2} F^a \wedge \star F^a - \frac{1}{2e^2} D\Phi \wedge \star D\Phi - V(\Phi) \star 1$$

and the equations are

$$\begin{aligned} -d \star F - A \wedge \star F + \star F \wedge A - [\Phi, \star D\Phi] &= 0 \,, \\ d \star D\Phi + A \wedge \star D\Phi + \star D\Phi \wedge A - \lambda \left(\Phi^a \Phi^a - \nu^2\right) \star 1 &= 0 \,. \end{aligned}$$

We will compute the equations using Mathematica and the package RGTC in a few lines for the ansatz

$$A = -\frac{W(t)}{\sqrt{2}} \left(\mathrm{t}_1 \rho \mathrm{d}\varphi - \mathrm{t}_2 \mathrm{d}\rho\right) - \frac{1}{2} \mathrm{t}_3 \mathrm{d}\varphi \,, \qquad \Phi = G(t)\, \mathrm{t}_3 \,.$$

Easy way to evaluate the equation with the computer

```
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```

```
t1 = I PauliMatrix[1]; t2 = I PauliMatrix[2]; t3 = I PauliMatrix[3]; id2 = IdentityMatrix[2];
```

```
A = -W[t]/Sqrt[2] (t1 ρ d[ϕ] - t2 d[ρ]) - 1/2 t3 d[ϕ]; F = d[A] + A ∧ A; Φs = G[t] t3;
```

```
com[x_, y_] := x.y - y.x; DΦ = d[Φs] + com[A, Φs];
```

$$A = -\frac{W(t)}{\sqrt{2}}\left(\mathrm{t}_1\rho\mathrm{d}\varphi - \mathrm{t}_2\mathrm{d}\rho\right) - \frac{1}{2}\mathrm{t}_3\mathrm{d}\varphi$$
$$\Phi = G\left(t\right)\mathrm{t}_3$$
$$D\Phi = d\Phi + \left[A, \Phi\right]$$

$$\sqrt{2}$$

$$\left\{\frac{i\,(e[1]\wedge e[2]\wedge e[4] + i\,e[1]\wedge e[3]\wedge e[4])\,\left(4\,G[t]^2\,W[t] + 2\,W[t]^3 + W''[t]\right)}{\sqrt{2}},\ 0\right\}\Big\}$$

```
Collect[Simplify[d[HStar[DΦ]] + A ∧ HStar[DΦ] + HStar[DΦ] ∧ A - HStar[1] λ (-1/2 Tr[Φs.Φs] - v^2) Φs],
 e[1] ∧ e[2] ∧ e[3] ∧ e[4]]
```

$$\left\{\left\{-i\,e[1]\wedge e[2]\wedge e[3]\wedge e[4]\,\left(\lambda\,G[t]^3 + G[t]\,\left(-\lambda\,v^2 + 4\,W[t]^2\right) + G''[t]\right),\ 0\right\},\ \left\{0,\ i\,e[1]\wedge e[2]\wedge e[3]\wedge e[4]\,\left(\lambda\,G[t]^3 + G[t]\,\left(-\lambda\,v^2 + 4\,W[t]^2\right) + G''[t]\right)\right\}\right\}$$

```
<< EDCRGTCcode`
```

$$
\begin{aligned}
-d\star F - A\wedge\star F + \star F\wedge A - [\Phi,\star D\Phi] &= 0\,,\\
d\star D\Phi + A\wedge\star D\Phi + \star D\Phi\wedge A - \lambda\left(\Phi^a\Phi^a - \nu^2\right)\star 1 &= 0\,.
\end{aligned}
$$

```
eIN = {d[t], d[ρ], ρ d[ϕ], d[z]}; xIN = {t, ρ, ϕ, z}; gIN = DiagonalMatrix[{-1, 1, 1, 1}];
RGtensors[ gIN, xIN, eIN, {0, 0}] // Simplify ;
```

```
Simplify[-d[HStar[F]] - A∧HStar[F] + HStar[F]∧A - com[Φs, HStar[DΦ]]]
```

$$
\left\{\left\{0,\ \frac{(i\,e[1]\wedge e[2]\wedge e[4] + e[1]\wedge e[3]\wedge e[4])\left(4\,G[t]^2\,W[t] + 2\,W[t]^3 + W''[t]\right)}{\sqrt{2}}\right\},\ \left\{\frac{i\,(e[1]\wedge e[2]\wedge e[4] + i\,e[1]\wedge e[3]\wedge e[4])\left(4\,G[t]^2\,W[t] + 2\,W[t]^3 + W''[t]\right)}{\sqrt{2}},\ 0\right\}\right\}
$$

```
Collect[Simplify[d[HStar[DΦ]] + A∧HStar[DΦ] + HStar[DΦ]∧A - HStar[1] λ (-1/2 Tr[Φs.Φs] - ν^2 ) Φs],
 e[1]∧e[2]∧e[3]∧e[4]]
```

$$
\left\{\left\{-i\,e[1]\wedge e[2]\wedge e[3]\wedge e[4]\left(\lambda\,G[t]^3 + G[t]\left(-\lambda\,\nu^2 + 4\,W[t]^2\right) + G''[t]\right),\ 0\right\},\ \left\{0,\ i\,e[1]\wedge e[2]\wedge e[3]\wedge e[4]\left(\lambda\,G[t]^3 + G[t]\left(-\lambda\,\nu^2 + 4\,W[t]^2\right) + G''[t]\right)\right\}\right\}
$$

The reduced Newtonian system

The important property of the ansatz for the fields is that the full coupled system of non-linear partial differential equations reduces to

two coupled ordinary differential equations

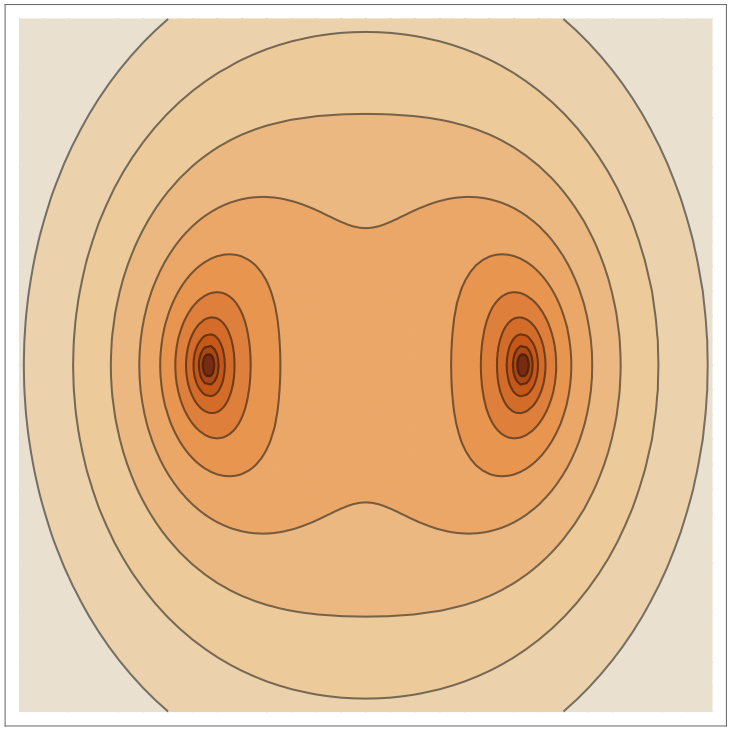
$$\begin{aligned} \frac{\mathrm{d}^2 G}{\mathrm{d}t^2} + 4W^2 G + \lambda \left(G^2 - \nu^2\right) G &= 0 \\ \frac{\mathrm{d}^2 W}{\mathrm{d}t^2} + 4G^2 W + 2W^3 &= 0 \end{aligned}$$

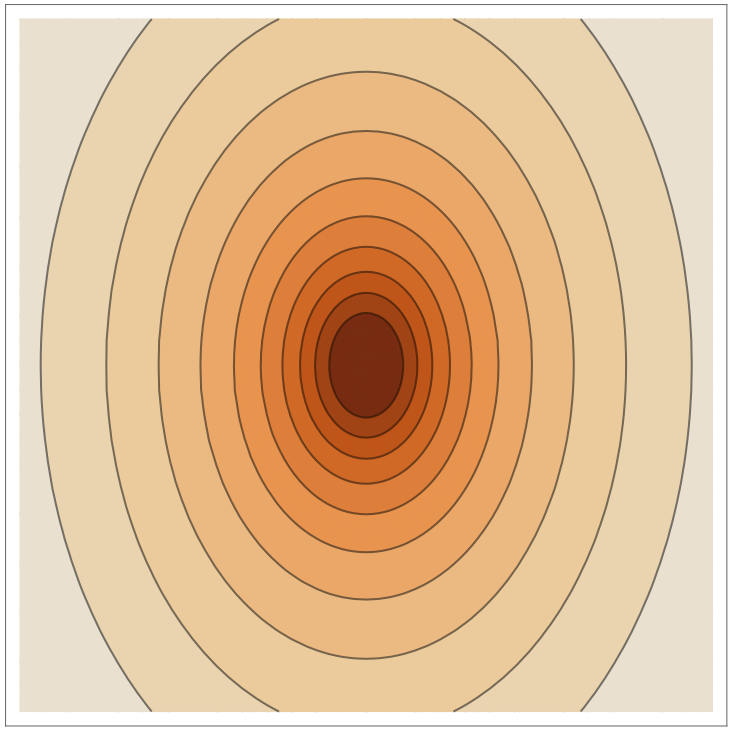
The reduced Newtonian system

It is conceptually useful to rewrite this system as a Newtonian system for the variables (G, W)

$$\frac{\mathrm{d}^2 G}{\mathrm{d}t^2} = -\frac{\partial}{\partial G} V(G, W) \ , \qquad \frac{\mathrm{d}^2 W}{\mathrm{d}t^2} = -\frac{\partial}{\partial W} V(G, W)$$

$$V(G, W) = 2W^2 G^2 + \frac{1}{2} W^4 + \frac{\lambda}{4} \left(G^2 - \nu^2\right)^2$$





Properties of effective system

- Integrating the Newtonian system once we find that the energy density $\mathcal{E}$ is conserved, in spite of the fact that the fields are time dependent. This is consistent with the absence of energy fluxes.

One could conclude then that this configuration is static and hence there is no characteristic frequencies of the system

$$\Downarrow$$

This idea will lead us to study a scalar probe on this background

- **Shift symmetry**

$$t \to t - t_0$$

- **Scaling invariance**

$$(t, W, G, \nu, \lambda) \to \left(\frac{t}{T}, TW, TG, T\nu, \lambda\right)$$

Analytic Solutions

Map of the analytic solutions

Without VEV
$\nu = 0$

With VEV
$\nu \neq 0$

Perturbed solution
$W = 0\,, G = 0$

Perturbed solution
$W = 0\,, G = \pm\nu$

Pure Yang-Mills (a, t_0)
$W(t) = \pm a\ \mathrm{sn}(at, -1)$

Pure Higgs (a, t_0)
$G(t) = \pm\sqrt{2} a\, \mathrm{sn}\left(a\sqrt{\lambda} t, -1\right)$

Pure Higgs (q, t_0)
$G(t) = \pm\nu\sqrt{\frac{-2q}{1+q}}\, \mathrm{sc}\left(-\sqrt{\frac{\lambda}{1+q}}\, \nu\, t, 1-q\right)$

Both fields (a, t_0)
$W(t) = \pm a\ \mathrm{sn}\left(a\sqrt{\frac{8-\lambda}{4-\lambda}} t, -1\right)$
$G(t) = \pm\sqrt{\frac{2}{4-\lambda}}\, a\, \mathrm{sn}\left(a\sqrt{\frac{8-\lambda}{4-\lambda}} t, -1\right)$

Both fields (t_0)
$G(t) = \pm_1\sqrt{2}\, \nu\, \mathrm{dn}\left(\sqrt{8-\lambda}\, \nu t -, \frac{\lambda}{8-\lambda}\right)$
$W(t) = \pm_2\sqrt{\frac{\lambda(\lambda-4)}{8-\lambda}}\, \nu\, \mathrm{sn}\left(\sqrt{8-\lambda}\, \nu t, \frac{\lambda}{8-\lambda}\right)$

Is the Yang-Mills-Higgs system in this ansatz integrable?

The ansatz reduced the system to an effective 2D Newtonian system. The variation w.r.t. the fields of the on-shell action leads the correct equations

$$\boxed{\begin{array}{l} A = -\frac{W(t)}{\sqrt{2}}\left(\mathrm{t}_1 \rho \mathrm{d}\varphi - \mathrm{t}_2 \mathrm{d}\rho\right) - \frac{1}{2}\mathrm{t}_3 d\varphi \\ \Phi = G\left(t\right)\mathrm{t}_3 \end{array}} \rightarrow L = \frac{1}{2}\dot{W} + \frac{1}{2}\dot{G} - V\left(W, G\right)$$

Going to the Hamiltonian formalism defined on a phase space $\mathbf{P}$ with Hamiltonian

$$H = \frac{1}{2}p_W^2 + \frac{1}{2}p_G^2 + 2W^2G^2 + \frac{1}{2}W^4 + \frac{\lambda}{4}\left(G^2 - \nu^2\right)^2$$

We will integrate numerically the system and study the properties. In particular the **transition to chaos**.

Both fields $\nu = 0$ case

In this talk we will restrict ourself to analyze the vanishing VEV case $\nu = 0$ with both fields turned on

$$G(t;a) = \pm\sqrt{\frac{2}{4-\lambda}}a\,\mathrm{sn}\left(a\sqrt{\frac{8-\lambda}{4-\lambda}}(t-t_0),-1\right),$$

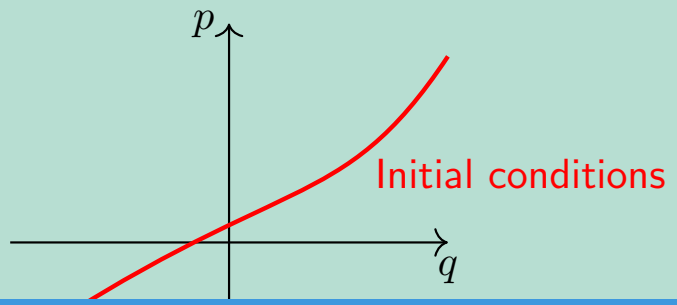
$$W(t;a) = \pm a\,\mathrm{sn}\left(a\sqrt{\frac{8-\lambda}{4-\lambda}}(t-t_0),-1\right).$$

This solution is explicitly real for $\lambda < 4$ and $a \in \mathbb{R}$.

$$\mathcal{E} = \tfrac{1}{2}a^4(\lambda-8)(\lambda-6)(\lambda-4)^{-2}$$

This solution will define a **set of initial conditions** that lead periodic integrable solutions:

$$\begin{aligned} X_a : \mathbb{R} &\to \mathbf{P} \\ a &\mapsto [G, W, p_G, p_W](0,a) \end{aligned}$$



Exploring the phase space beyond the integrable sector

Diagnosing Chaos

A chaotic behavior is usually defined as the sensibility of the initial conditions. Chaos is often quantified using Lyapunov exponents.

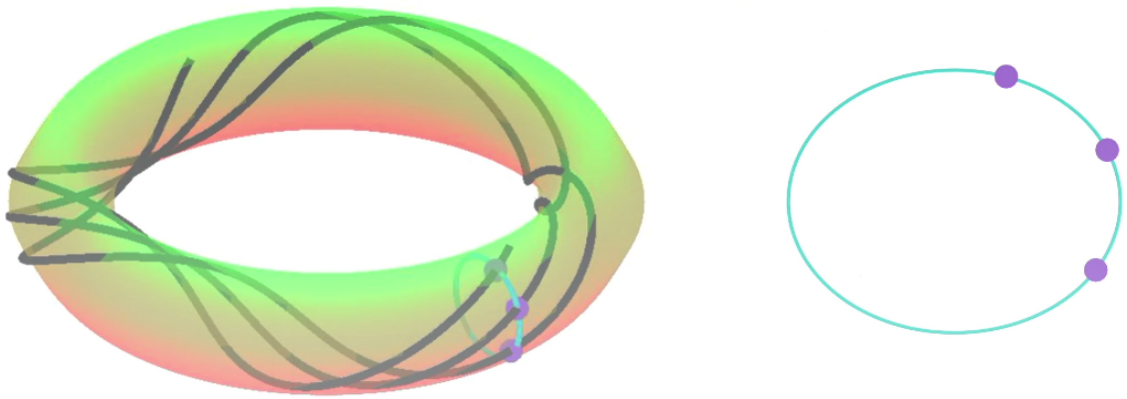
To diagnose chaos we will use the following indicators

- Poincaré sections
- Fourier spectra
- Deviation of near-by trajectories in the phase space

Diagnosing Chaos: Poincaré sections

We define the phase space P as four dimensional manifold with coordinates (W, G, p_W, p_G) . Where the dynamics is generated by the Hamiltonian H through the Poisson bracket $\dot{X} = [X, H]$.

Poincaré section is sub-manifold $\Sigma \subset P$ which intersect the trajectory of the trajectory in phase space[1]



[1] Picture from Jason Bramburger YouTube channel.

Diagnosing Chaos: Geodesic deviation

How to measure the deviation of near-by trajectories in phase space?

Differential geometry Given a manifold with a metric $g_{\mu\nu}$ in coordinates x^μ, the geodesics satisfy

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu{}_{\rho\sigma} \frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda} \frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda} = 0$$

The deviation of the near-by geodesics is captured by the Jacobi-Levi-Civita equation for the Jacobi field η^μ

$$\nabla_\sigma \nabla^\sigma \eta^\mu - R^\mu{}_{\rho\sigma\lambda} \frac{\mathrm{d}x^\rho}{\mathrm{d}\lambda} \frac{\mathrm{d}x^\sigma}{\mathrm{d}\lambda} \eta^\lambda = 0 \ .$$

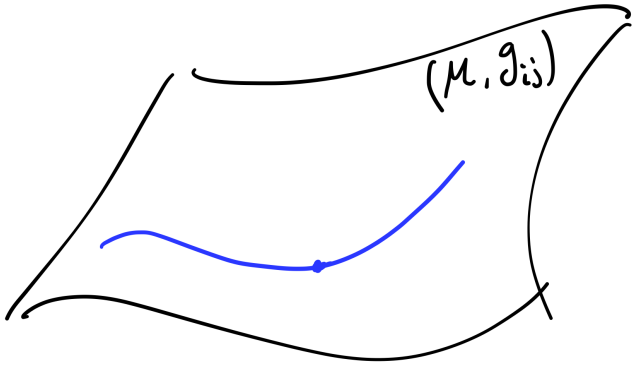
We will use this tools in our effective Newtonian system

Diagnosing Chaos: Geodesic deviation

Let us consider a Newtonian system with generalized coordiantes q^i and potential $V(q)$

Definition (Jacobi metric) We define the Jacobi metric of a manifold with coordinates q^i as

$$ds^2 = 2\left[E - V(q)\right]\delta_{ij}dq^i q^j$$



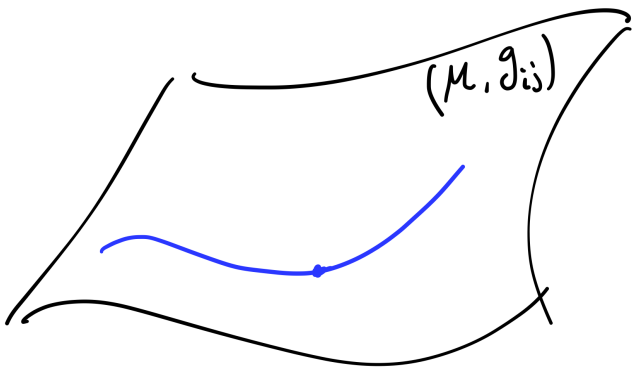
Diagnosing Chaos: Geodesic deviation

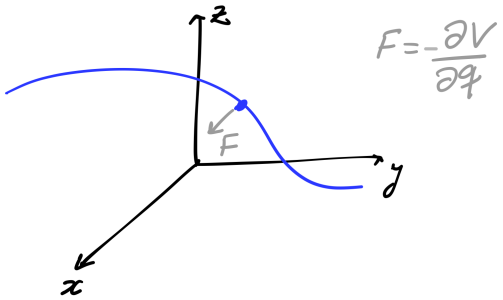
One benefit of this metric is that the geodesic equation on this manifold are Newton's equations[2]

$$\frac{\mathrm{d}^2 q^i}{\mathrm{d}\lambda^2} + \Gamma^i{}_{jk} \frac{\mathrm{d}q^j}{\mathrm{d}\lambda} \frac{\mathrm{d}q^k}{\mathrm{d}\lambda} = 0$$

$$\Longleftrightarrow$$

$$\frac{\mathrm{d}^2 q^i}{\mathrm{d}t} = -\delta^{ij} \frac{\partial V}{\partial q^j}$$





[2] $\Gamma^i{}_{jk} = \delta^i_{(k}\partial_{j)} W - \frac{1}{2}\delta_{jk}\partial^i \log W$, $\quad \frac{d\lambda}{dt} = E - V(q)$

Diagnosing Chaos: Geodesic deviation

Let us go back to our problem: We specialize for the case of 2 degree of freedom $\left(q^1, q^2\right) = \left(G\left(t\right), W\left(t\right)\right)$ ['96 Cerruti, Pettini]

$$\nabla_\sigma \nabla^\sigma \eta^i - \mathcal{R}^i{}_{jkl} \frac{\mathrm{d}q^j}{\mathrm{d}\lambda} \frac{\mathrm{d}q^k}{\mathrm{d}\lambda} \eta^l = 0$$

To evaluate the Jacobi-Levi-Civita equation we use the fact that $\frac{d^2}{12}\left(d^2 - 1\right)\Big|_{d=2} = 1$

$$\mathcal{R}^{ij}{}_{kl} = \frac{1}{2}\mathcal{R}\left(\delta^i_k \delta^j_l - \delta^i_l \delta^j_k\right)$$

Then, the Jacobi-Levi-Civita equation reduced to

$$\nabla^2 \eta^i + \frac{\mathcal{R}}{2}\left(\eta^i - \frac{dq^i}{d\lambda}\frac{dq^j}{d\lambda}\eta_j\right) = 0$$

Diagnosing Chaos: Geodesic deviation

In 2 dimensions there exist ε_{ij} then we contract with $\frac{dq_i}{d\lambda}$ and $\varepsilon_{ij}\frac{dq_j}{d\lambda}$ leading to

$$\frac{d^2\eta_\perp}{d\lambda^2} + \frac{\mathcal{R}}{2}\eta_\perp = 0 \ , \qquad \frac{d^2\eta_\parallel}{d\lambda^2} = 0$$

Observations

- The system gets decoupled.
- Negative scalar curvature would lead to solutions with exponential grow in time for $\eta_\perp$.

In our case (in most cases) $R > 0 \implies$ Instabilities should come from parametric resonances, as $R\,(t)$

$$R = \frac{1}{2(\mathcal{E}-V)^3}\left[4W^2\left(2G^2+W^2\right)^2 + G^2\left(4W^2+\lambda\left(G^2-\nu^2\right)\right)^2\right] + \frac{1}{2(\mathcal{E}-V)^2}\left[(4+3\lambda)G^2 + 10W^2 - \lambda\nu^2\right].$$

Diagnosing Chaos: Geodesic deviation

Going back to Newtonian time and defining $Y = \eta_\perp / \sqrt{\mathcal{E} - V}$

$$\frac{\mathrm{d}^2 Y}{\mathrm{d}t^2} + \Sigma\left(t\right) Y = 0 \ ,$$

where

$$\Sigma = 2\mathcal{R}\left(\mathcal{E} - V\right)^2 - \frac{1}{2\left(\mathcal{E} - V\right)} \frac{\mathrm{d}^2 V}{\mathrm{d}t^2} - \frac{3}{4\left(\mathcal{E} - V\right)^2} \left(\frac{\mathrm{d}V}{\mathrm{d}t}\right)^2$$

This is the equation that we will integrate to detect a exponential growing separation of the near-by geodesics.

The strategy

We will parametrize the initial conditions as keeping free the energy

$$
\begin{aligned}
q_1 &= G\left(t; a\right) \\
q_2 &= W\left(t; a\right) \\
p_1 &= \sqrt{2\left[\mathcal{E} - V\left(G, W\right)\right] - p_2^2}\ , \\
p_2 &= \dot{W}\left(t; a\right)
\end{aligned}
$$

In the most general case with two non-trivial fields dynamics

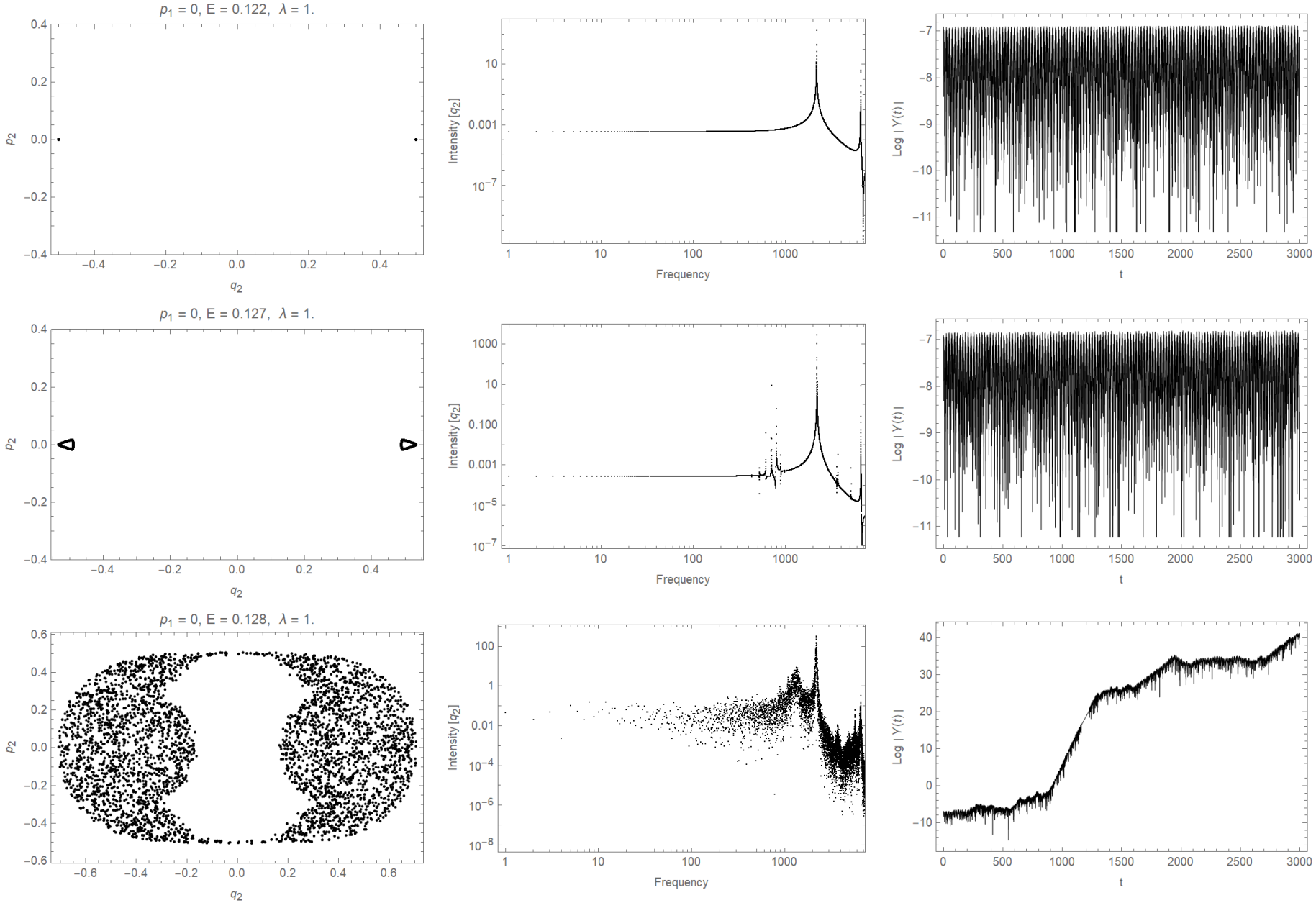
- Case without VEV

$$
\left(a, \mathcal{E}\right) : \mathbb{R} \times \left[\mathcal{E}_0, \infty\right[\to \mathbf{P}\ , \qquad \text{where} \qquad \mathcal{E}_0 = \frac{a^4\left(\lambda - 8\right)\left(\lambda - 6\right)}{4\left(\lambda - 4\right)^2}
$$

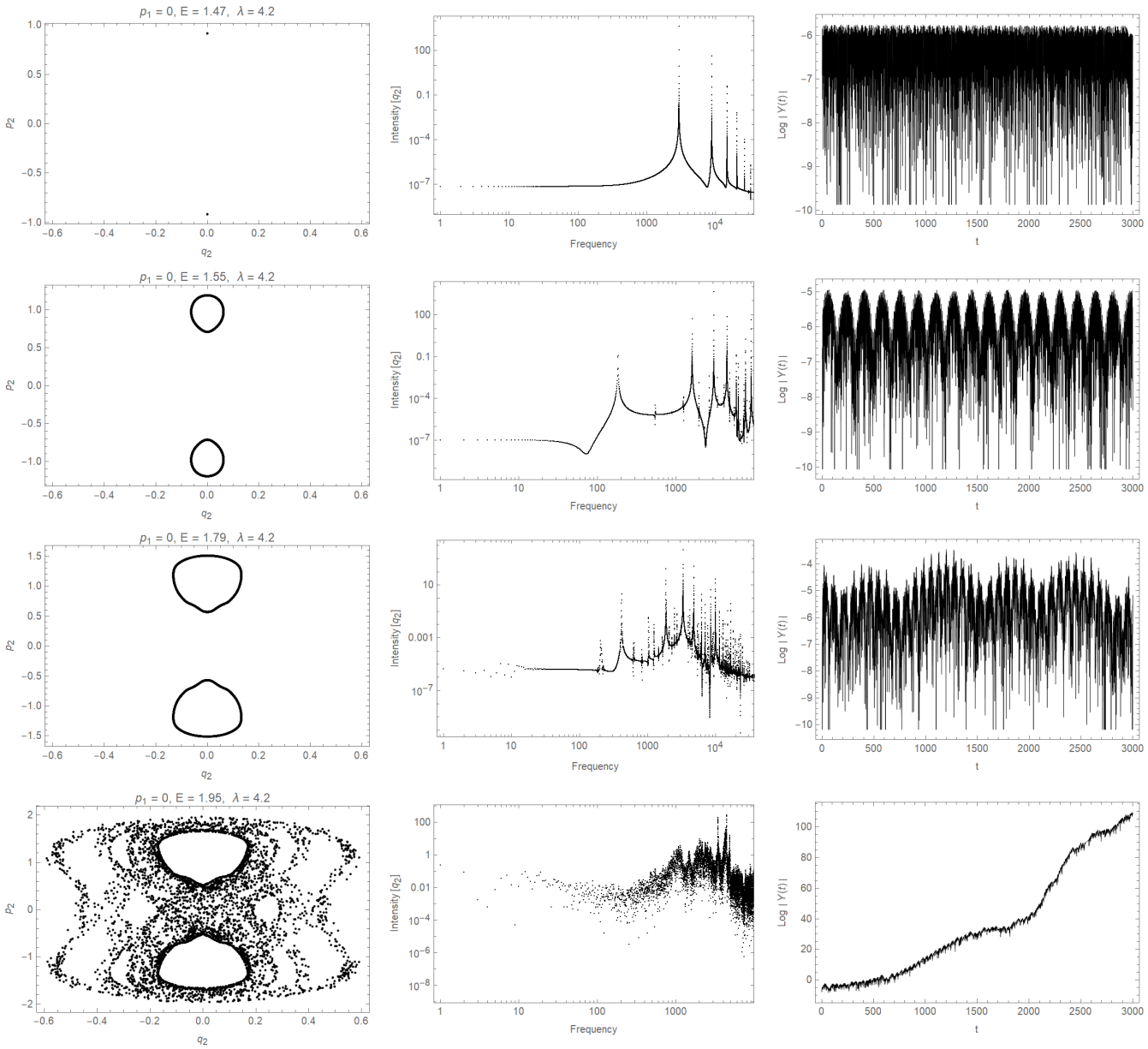
- In the case with VEV we have only the

$$
\left(\mathcal{E}\right) : \left[\mathcal{E}_0, \infty\right[\to \mathbf{P}\ , \qquad \text{where} \qquad \mathcal{E}_0 = \frac{\lambda\nu^4}{4}\left(2\lambda - 7\right)
$$

Case without VEV for $a = 0.5$



Case with VEV for $\nu = 1$

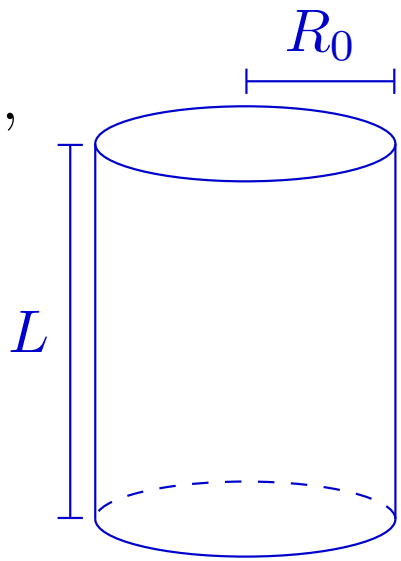


What about the resonances? Let us put a probe scalar field

Probe scalar field

To test the possible resonances we consider a scalar fields in the adjoint of $SU(2)$ coupled to the non-Abelian fields

$$
\begin{aligned}
S &= -\int dt \int d^3x \sqrt{\gamma} D_\mu \phi^\dagger D^\mu \phi \ , \\
D_\mu \phi &= \partial_\mu \phi + A_\mu \phi
\end{aligned}
$$



in the background

$$
\begin{aligned}
\gamma_{ij} dx^i dx^j &= \mathrm{d}z^2 + \mathrm{d}\rho^2 + \rho^2 \mathrm{d}\varphi^2 \ , \\
A &= -\frac{W(t)}{\sqrt{2}} \left(\mathrm{t}_1 \rho \mathrm{d}\varphi - \mathrm{t}_2 \mathrm{d}\rho \right) - \frac{1}{2} \mathrm{t}_3 \mathrm{d}\varphi \ .
\end{aligned}
$$

The momentas are

$$
p_a(x) = \frac{\delta L}{\delta \partial_t \Phi^a(x)} = \sqrt{\gamma} \dot{\phi}^*{}_a \ , \qquad \tilde{p}^a(x) = \frac{\delta L}{\delta \partial_t \Phi^*{}_a(x)} = \sqrt{\gamma} \dot{\phi}^{*a} \ .
$$

Probe scalar field

We separate the Hamiltonian as

$$H = H_0 + H_I$$

$$\begin{aligned} H_0 &= \int d^3x \left[\frac{p_a \tilde{p}^a}{\sqrt{\gamma}} + \sqrt{\gamma}\gamma^{ij}\partial_i \phi^\dagger \partial_j \phi - \phi^\dagger \sqrt{\gamma} \left(-\frac{1}{\rho^2} t_3 \partial_\varphi - \frac{1}{4\rho^2} \right) \phi \right] \\ H_{int} &= -\sqrt{2} W(t) \int d^3x \phi^\dagger \sqrt{\gamma} \left(\frac{1}{2\rho} t_2 + t_2 \partial_\rho - \frac{1}{\rho} t_1 \partial_\varphi - \frac{W}{\sqrt{2}} \right) \phi \end{aligned}$$

Probe scalar field

The free equation for the scalar probe is

$$-\partial_t^2\boldsymbol{\phi}+\partial_z^2\boldsymbol{\phi}+\frac{1}{\rho}\partial_\rho\boldsymbol{\phi}+\partial_\rho^2\boldsymbol{\phi}+\frac{1}{\rho^2}\partial_\varphi^2\boldsymbol{\phi}-\frac{1}{4\rho^2}\boldsymbol{\phi}-\frac{1}{\rho^2}t_3\partial_\varphi\boldsymbol{\phi}=\mathbf{0}$$

which is solved by $\boldsymbol{\phi}=\bar{\phi}+\phi$

$$\begin{aligned}
\bar{\phi} &= \bar{\sum_{\ell mn}}\frac{\bar{\mathcal{N}}_{\ell mn}}{\sqrt{2\bar{\omega}_{\ell mn}}}\sin\left(\frac{\pi\ell z}{L}\right)\\
&\quad\times J_{m-\frac{1}{2}}\left(\frac{\alpha_n^{m-\frac{1}{2}}\rho}{R_0}\right)\begin{pmatrix}\boldsymbol{l}_{\uparrow\ell mn}e^{i(m\varphi-\bar{\omega}_{\ell mn}t)}+\tilde{\boldsymbol{l}}^\dagger_{\uparrow\ell mn}e^{i(m\varphi+\bar{\omega}_{\ell mn}t)}\\ \boldsymbol{l}_{\downarrow\ell mn}e^{-i(m\varphi+\bar{\omega}_{\ell mn}t)}+\tilde{\boldsymbol{l}}^\dagger_{\downarrow\ell mn}e^{-i(m\varphi-\bar{\omega}_{\ell mn}t)}\end{pmatrix}\\
\phi &= \sum_{\ell mn}\frac{\mathcal{N}_{\ell mn}}{\sqrt{2\omega_{\ell mn}}}\sin\left(\frac{\pi\ell z}{L}\right)\\
&\quad\times J_{m+\frac{1}{2}}\left(\frac{\alpha_n^{m+\frac{1}{2}}\rho}{R_0}\right)\begin{pmatrix}\boldsymbol{r}_{\uparrow\ell mn}e^{-i(m\varphi+\omega_{\ell mn}t)}+\tilde{\boldsymbol{r}}^\dagger_{\uparrow\ell mn}e^{-i(m\varphi-\omega_{\ell mn}t)}\\ \boldsymbol{r}_{\downarrow\ell mn}e^{i(m\varphi-\omega_{\ell mn}t)}+\tilde{\boldsymbol{r}}^\dagger_{\downarrow\ell mn}e^{i\left(m\varphi+\omega'_{\ell mn}t\right)}\end{pmatrix}
\end{aligned}$$

Probe scalar field

$$\bar{\sum_{\ell mn}} = \sum_{\ell=1}^{\infty}\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}, \quad \sum_{\ell mn} = \sum_{\ell=1}^{\infty}\sum_{m=0}^{\infty}\sum_{n=1}^{\infty}, \quad \boldsymbol{\phi} = \bar{\phi} + \phi\ .$$

$$\bar{\mathcal{N}}_{\ell mn} = \sqrt{\frac{2}{\pi L}}\frac{1}{R_0 J_{m+\frac{1}{2}}\left(\alpha_n^{m-\frac{1}{2}}\right)}, \quad \mathcal{N}_{\ell mn} = \sqrt{\frac{2}{\pi L}}\frac{1}{R_0 J_{m+\frac{3}{2}}\left(\alpha_n^{m+\frac{1}{2}}\right)},$$

$$(\bar{\omega}_{\ell mn})^2 = \left(\frac{\alpha_n^{m-\frac{1}{2}}}{R_0}\right)^2 + \left(\frac{\pi\ell}{L}\right)^2, \qquad (\omega_{\ell mn})^2 = \left(\frac{\alpha_n^{m+\frac{1}{2}}}{R_0}\right)^2 + \left(\frac{\pi\ell}{L}\right)^2.$$

$$\begin{aligned}
\bar{\phi} &= \bar{\sum_{\ell mn}} \frac{\bar{\mathcal{N}}_{\ell mn}}{\sqrt{2\bar{\omega}_{\ell mn}}} \sin\left(\frac{\pi\ell z}{L}\right) \\
&\quad \times J_{m-\frac{1}{2}}\left(\frac{\alpha_n^{m-\frac{1}{2}}\rho}{R_0}\right)\begin{pmatrix} \boldsymbol{l}_{\uparrow\ell mn}e^{i(m\varphi-\bar{\omega}_{\ell mn}t)} + \tilde{\boldsymbol{l}}^{\dagger}_{\uparrow\ell mn}e^{i(m\varphi+\bar{\omega}_{\ell mn}t)} \\ \boldsymbol{l}_{\downarrow\ell mn}e^{-i(m\varphi+\bar{\omega}_{\ell mn}t)} + \tilde{\boldsymbol{l}}^{\dagger}_{\downarrow\ell mn}e^{-i(m\varphi-\bar{\omega}_{\ell mn}t)} \end{pmatrix} \\
\phi &= \sum_{\ell mn} \frac{\mathcal{N}_{\ell mn}}{\sqrt{2\omega_{\ell mn}}} \sin\left(\frac{\pi\ell z}{L}\right) \\
&\quad \times J_{m+\frac{1}{2}}\left(\frac{\alpha_n^{m+\frac{1}{2}}\rho}{R_0}\right)\begin{pmatrix} \boldsymbol{r}_{\uparrow\ell mn}e^{-i(m\varphi+\omega_{\ell mn}t)} + \tilde{\boldsymbol{r}}^{\dagger}_{\uparrow\ell mn}e^{-i(m\varphi-\omega_{\ell mn}t)} \\ \boldsymbol{r}_{\downarrow\ell mn}e^{i(m\varphi-\omega_{\ell mn}t)} + \tilde{\boldsymbol{r}}^{\dagger}_{\downarrow\ell mn}e^{i\left(m\varphi+\omega'_{\ell mn}t\right)} \end{pmatrix}
\end{aligned}$$

Probe scalar field

We compute the transition amplitude of a particular process

$$\langle 0| r_{\downarrow \ell' m' n'} H_{\mathsf{int}} \boldsymbol{l}^{\dagger}_{\uparrow \ell m n} |0\rangle$$
$$= -\frac{\pi L \mathcal{N}_{\ell' m' n'} \bar{\mathcal{N}}_{\ell m n}}{R_0 \sqrt{2\omega_{\ell' m' n'} \bar{\omega}_{\ell m n}}} \delta^{m'}_{m} \delta^{\ell'}_{\ell} \alpha_n^{m-\frac{1}{2}} \int dt\, W(t) e^{i(\omega_{\ell' m' n'} - \bar{\omega}_{\ell m n})t} \times$$
$$\times \int_0^{R_0} d\rho\, \rho\, J_{m'+\frac{1}{2}} \left(\alpha_{n'}^{m'+\frac{1}{2}} \frac{\rho}{R_0} \right) J_{m+\frac{1}{2}} \left(\alpha_n^{m-\frac{1}{2}} \frac{\rho}{R_0} \right)$$

We see that depending the form of the function $W(t)$ we can detect it in the transitions of a scalar.

Future works

We can use the tools that considered to diagnose chaos in this work to explore other systems

- Studying Four-Dimensional Stationary and Axially Symmetric Spacetimes: The system of equations coming from the geodesic equations can be reduced to a two-dimensional effective Newtonian problem.
- Explore more realistic models with similar techniques, such as Skyrme model, Sigma models, or Yang-Mills models with scalar fields in the fundamental representation.

Conclusions

- We explored a sector $SU(2)$ Georgi-Glashow model by considering ansatz for fields that preserve spatial dependence and topological fluxes, facilitating the analysis of chaos with varying parameters like Higgs coupling and vacuum expectation value.
- We identify an integrable sector, with exact solutions representing a non-Abelian alternating current generator.
- Increased energy leads to the loss of integrability, and chaotic behavior relates to parametric resonance shown by geodesic deviation growth in Jacobi metric.

Gracias!